Atem Overseer user guide

Atem Overseer monitors and controls a fleet of Blackmagic ATEM switchers from one screen: one tile per switcher, showing live preview, audio, record and stream state, with transport controls under each.

> **Before you rely on this:** it was developed and verified end to end against the built-in simulated fleet (`--mock`), with real test-pattern streams pushed into the RTMP ingest so the picture in the tiles is real even though the switcher state is simulated. It's **since been tested against real ATEM hardware in a lab**, but **never on a live show**.
>
> **There's also no authentication, and the server listens on every network interface.** Anyone who can reach port 4700 can start and stop recording and streaming on your switchers, with no password. See Security.
>
> This project was written with AI assistance and reviewed by a human. Use at your own risk in live environments.

Three other documents cover the subsystems in depth, and I haven't repeated them here: device-management.md (adding devices, discovery, launching external apps), streaming-setup.md (getting live preview working end to end) and restreamer.md (the optional split pipeline for multi-destination egress). This guide is the operator's overview.

Getting started

```
npm install
npm run dev:mock          # simulated 3-switcher fleet — dashboard at http://localhost:4700
```

For real devices, copy the example config, edit the addresses, then build and start:

```
cp atem-overseer.config.example.json atem-overseer.config.json
npm run build
npm start
```

There's also a one-click desktop app (see the README's Desktop app section).

`publicHost` is the setting that breaks everything if it's wrong

Set it to **the address the ATEMs and the browsers reach this machine at**. It's baked into the generated `Streaming.xml` and into the http-flv playback URLs.

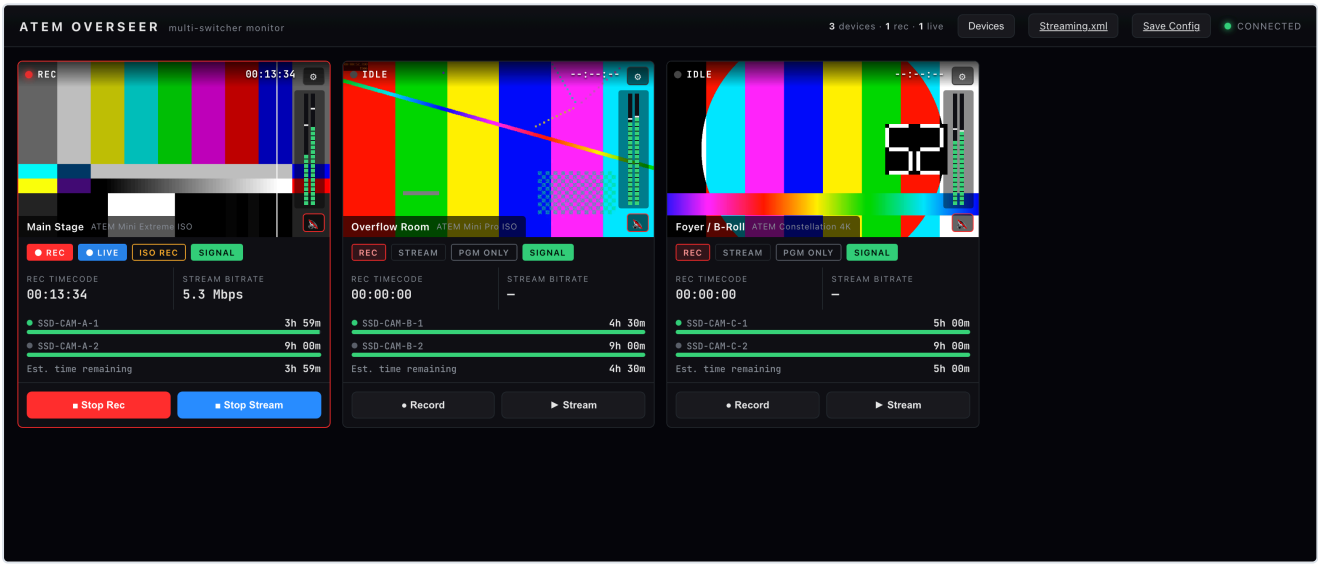
If it's `localhost`, or an interface the switchers can't route to, the switchers won't find the ingest and the browser won't find the playback, **and nothing will tell you that's why**. The tiles just stay dark.

A note on `--mock`

`--mock` simulates every switcher, so nothing touches hardware. But it only substitutes the built-in demo fleet **if your config file has no devices**. With a real config present you'll see **your own device names, simulated**. Don't take familiar names on screen as proof you're connected to anything.

Reading a tile

Each tile is one switcher.



The simulated fleet. The pictures are real test-pattern streams through the RTMP ingest; the switcher state behind them is `--mock`.

Most of a tile is self-explanatory. These five aren't:

What you see	What it actually is
Drive capacity bar	**Recording time remaining, not disk space.** The ATEM protocol doesn't expose capacity at all — only seconds of headroom. The bar uses **4 hours as a nominal "full"**, so a fresh 1 TB SSD does not read 100%.
Version	The **ATEM protocol/API version**, not the switcher's firmware. Firmware isn't available over the wire.
Audio meters	**Always live telemetry**, regardless of any mute.
Stream cache	Fraction of the switcher's stream buffer in use — a **network-health** indicator, not progress. A rising cache means the upstream is struggling.
Hostname	Reverse-DNS, often blank. Blank means DNS, not a fault.

Connection state is `connecting` / `connected` / `disconnected`, per device, and one switcher dropping doesn't affect the rest of the fleet.

The two different mutes

This catches everyone once:

- **The per-tile mute button** is **browser-local**. It silences *your* playback of that tile's stream. It changes nothing on the switcher, and **it doesn't affect the meters**: the meter keeps moving on a muted tile, on purpose.
- **Monitor mute** is the **ATEM's own monitor bus**, sent to the switcher. It affects what comes out of the switcher's monitor output, for everyone.

If someone says "I muted it and it's still loud in the room", they used the browser one.

Transport controls

Record start/stop, stream start/stop, and record mode (**PGM** or **ISO**).

These are **real commands to real switchers that may be live on air**. There's no confirmation step and no undo. Two consequences:

- **Check which tile you're on.** The dashboard is designed for a fleet, so the buttons are close together by nature.
- **Anyone who can reach the dashboard can press them** (see Security).

Recording writes to whatever disk the switcher has in its working set. Check the drive readout before a long record, bearing in mind it's *time*, not space.

Live preview

Full walkthrough in streaming-setup.md. The short version:

1. Tile ⚙ → **Download Streaming.xml**, and put it in ATEM Software Control's streaming support folder, or hit **Apply local service to switcher** to push the RTMP config directly over the protocol.
2. **Set the switcher's stream key to its Overseer device id.** This is the step people miss. The stream key is how the ingest knows which tile a feed belongs to; get it wrong and the stream arrives and lands nowhere.
3. Start streaming. The feed appears in the tile.

"Apply local service to switcher" doesn't work on every model. If the switcher's runner doesn't support remote streaming config you'll get `device does not support remote streaming config`. That's a limit of the switcher, not a bug. Use the XML route instead.

Managing the fleet

See device-management.md for discovery, adding devices by hand, and the external-app launch buttons. Two things worth repeating here:

- **The "launch external app" buttons run the app on the machine running the server**, not on the machine whose browser you're using. If you're viewing the dashboard remotely, ATEM Software Control opens at the server, where nobody's looking at it.
- **Importing an Overseer config XML saves it but does not apply device changes until restart.** The response says so, and the dashboard keeps showing the old fleet until you restart. This XML is Overseer's own fleet/ingest config. **It isn't an ATEM state backup**, and restoring it won't restore anything about the switchers.

Media pool

Behind the tile's gear: list the media pool, assign a media player to a slot, and upload a still.

Stills are converted in the browser to the switcher's resolution before upload. The upload cap is **64 MB**.

Transport, streaming and media upload are the paths where a simulator is least likely to match a real switcher, so test an upload against your own model before you need it on a show.

Security

The server has **no authentication and listens on every network interface**. Anyone who can reach port 4700 can **start and stop recording and streaming on your switchers**. There's no token, no login and no TLS, and you can't turn any of them on.

This belongs on a private production network: never on a venue guest network, and never port-forwarded.

Troubleshooting

Symptom	Cause
Tiles connect but preview stays dark	`publicHost` is wrong or unreachable, or the switcher's stream key isn't its Overseer device id.
A stream arrives but no tile shows it	Stream key ≠ device id.
"device does not support remote streaming config"	That model can't be configured remotely. Use `Streaming.xml`.
Drive bar looks wrong for the disk fitted	It is time remaining scaled against a nominal 4 hours, not capacity.
Muted a tile, still audible in the room	That is the browser mute; you want monitor mute.
Meters still moving on a muted tile	Intended — metering is telemetry and is always shown.
Imported a config, nothing changed	Device changes need a restart.
Launched ATEM Software Control, nothing appeared	It opened on the server's machine.
A command "worked" but nothing happened	Successful commands are not acknowledged; only failures produce a toast. Watch the tile state, not the button.
Sent `{"action":"begin"}` and it stopped	Anything that isn't exactly `start` means stop. There is no validation — see API.md.
Restreamer destinations vanished	A malformed `destinations` body is treated as an empty array and clears them — see API.md.
Phantom type errors when building	`build:libs` didn't run first — see DEVELOPING.md.
Desktop app's helpers die silently on macOS	Only on a self-built or pre-notarisation copy — the released `.pkg` builds are signed and notarised. Approving an unsigned `.app` does not unquarantine its bundled binaries. See the launcher README.

See also

- API.md: REST, WebSocket, snapshot fields, config schema
- DEVELOPING.md: the monorepo and the mock-first rule
- device-management.md · streaming-setup.md · restreamer.md